Does the choice of Sleep Regularity Index calculator or sleep/wake input change the conclusions? A re-analysis of the NHANES 2011–2014 sleep regularity–obesity association

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Abstract

Background. Open-source packages for the Sleep Regularity Index (SRI) can score the same recording differently. An earlier study reported an inverse cross-sectional association between the SRI and body mass index (BMI) in NHANES 2011–2014 from the released sleep/wake classification. The same data were re-analysed to examine whether that association depends on the calculator, the structure of the released classification or the derivation from the raw signal.

Methods. The released classification of the published sample ($n = 7085$) was rebuilt, four smoothed versions were derived, and the Phillips formula, `GGIR` and `sleepreg` were applied to each. The survey-weighted log-BMI models were refitted and the calculators tested on synthetic recordings. For a random sample of 1000 participants the series was derived anew from the 80 Hz raw signal with `GGIR` and the public SWaN code.

Results. The released classification switched between sleep and wake a median of 16 times per day. Smoothing raised the median SRI from 61 to 68–76. On the same input the calculators agreed to 0.1 points. The fully adjusted difference in BMI between the top and bottom SRI quintile was -4.5% for the released series and -2.8% to -3.7% after smoothing. In the random sample, SWaN rerun on the raw signal switched state 3.0 times per day against 17.8 for the release. The derived series scored 10.1 to 17.7 points higher. The same difference in BMI was -4.8% for the released classification and -0.7% to $+0.4\%$ for the derived series. Deleting one day of a synthetic recording (SRI 68.75) instead of marking it as missing made `sleepreg` return 35.18 without warning. Its default minimum of five valid days left 48% of participants unscored.

Conclusions. The SRI level depended on the input and only marginally on the calculator. The released classification is more fragmented than a new classification of the same signal with the algorithm's public code. The inverse association with BMI weakened with every smoothed and newly derived input. Reports should describe the derivation and smoothing of the series, the treatment of gaps, the minimum-day rule and the remaining sample, and archive the code.

Keywords: sleep regularity index; NHANES; accelerometry; wearable monitors; `GGIR`; `sleepreg`;

measurement error; reproducibility; missing data

1 Background

The Sleep Regularity Index [1] (SRI) measures the day-to-day similarity of a person's sleep/wake pattern. For every pair of epochs exactly 24 h apart it records whether the state is the same. The index is -100 plus 200 times the proportion of agreeing pairs. Identical days score 100 and a random pattern approximately 0 [2].

Czeisler et al. [20] compared the two most widely used open-source calculators, `sleepreg` [3] and `GGIR` [4, 5], in 73,794 adults of the UK Biobank. The packages gave markedly different scores and, through the calculator alone, different prospective associations with mortality, type 2 diabetes and atrial fibrillation. The authors proposed the Reporting Items for Regularity Indices (RIRI).

An earlier study [6] reported an inverse cross-sectional association between the SRI and body mass index (BMI) in the National Health and Nutrition Examination Survey (NHANES) 2011–2014 ($n = 7085$), more clearly in women. It computed the SRI with its own implementation of the Phillips formula from the minute-level sleep/wake classification that NHANES distributes with its accelerometer data (the variable `PAXPREDM`). Its results therefore rest on one calculator and one input.

Reported SRI values for the same NHANES accelerometry include a median of 78.1 [7], a mean of 63.0 [8] and the median of 63.9 (unweighted 61.5) of the earlier study. Cohorts whose epoch series come from a sleep-period detection step report medians of 81 [3] and 74.7 [9], whereas the NHANES classification is released minute by minute without such a step [10]. An SRI pipeline has two steps that can move the score, the derivation of the epoch-level series from the raw signal and the computation of the index from that series. Czeisler et al. [20] held the first step fixed. A released classification allows the complementary design, in which the same series is smoothed in defined ways and every version is passed to every calculator. NHANES has also released the raw 80 Hz signal [11], so that the first step can be repeated, with the sleep detection of `GGIR` [5, 12] and with the public code of SWaN (sleep-wear, wake-wear and nonwear classifier), the algorithm that its developers report to have generated the released classification [10].

The re-analysis therefore asked whether the published values can be rebuilt from the public files,

whether the structure of the released classification sets the level of the SRI, whether the calculators differ once they receive the same series, whether the published SRI–BMI association keeps its direction and size under smoothed and under newly derived series, and how a recording gap passed to a package changes its result.

2 Methods

2.1 Rebuilding the sleep/wake series

The minute-level sleep/wake series of the earlier study was rebuilt from the NHANES 2011–2014 minute files. `PAXPREDM` was recoded to sleep, wake or missing (non-wear and unknown). The published inclusion filters were then applied: full-minute observations, no quality flags, at least 3 days with less than 120 min of non-wear, days of exactly 1440 minutes, at least 3 consecutive days including a weekend day, and less than 30% missing minutes per participant. The series of each participant was then laid out as one continuous timeline with one value per minute from monitor placement to the end of the last retained day, every day starting at 00:00. Minutes and days that the filters had removed stayed in this timeline as missing values and were not deleted. Neither package checks whether successive days of a series are consecutive in time. When a day is deleted, `GGIR` pairs the date before the gap with the next date present, `sleepreg` carries the last state over the gap, and neither gives a warning (Section 3.4). A day marked as missing is left out of the comparison. The Phillips formula on this shared timeline of the 7085 participants (it returns a value for 7076 of them) was the same-input baseline. Its relation to the published values is described in Supplementary Section 3.

2.2 SRI calculators

The archived implementation of the earlier study computes the Phillips formula, $\mathrm{SRI} = -100 + \frac{200}{N_v}\sum 1[s_i = s_{i+1440}]$, at 1-min epochs over the N_v valid epoch pairs of consecutive days paired by

weekday. The SRI was recomputed on the rebuilt series with `sleepreg` 1.3.5 and `GGIR` 3.3-6 at their default settings. `GGIR` distinguishes SRI1, computed from the epoch-level sleep classification, from SRI2, computed from its own detected sleep period. The function `CalcSleepRegularityIndex`, the engine of SRI1, was called with its smoothing options unset. It returns a Phillips SRI for each pair of consecutive calendar days. The day-pair values were averaged where at least 66% of the epoch pairs were valid, which approximates the package's own criterion of more than 16 of 24 hours valid (`includenightcrit`). `sleepreg` re-expresses the series at 10-second epochs, drops noon-to-noon days with more than 6 h missing and returns no value below five valid comparison days (`minSRIdays`, also run at three). `pyActigraphy` [13] served as a fourth implementation on the synthetic recordings and on the 300 NHANES recordings with the fewest missing minutes (non-wear or unknown, Supplementary Table S2 and Section 2).

2.3 Epoch-level structure of the released classification and smoothed input variants

The released classification was described at the epoch level by the transitions between sleep and wake per day, the length of sleep bouts, the longest bout per day and the share of sleep minutes by clock hour. To separate the input from the calculator, four smoothed versions of the same series were derived and every calculator was run on each. The first is a centred rolling majority over 61 minutes (30 minutes on either side), computed over the observed minutes of the window. The second is the optional smoothing rule of `GGIR`'s SRI1 with the example values of its documentation, the same window with sleep assigned if at least 80% of it is sleep. The third bridges awakenings shorter than 30 minutes, after the WASO rule of `GGIR`'s SRI2 (`SRI2_WASOmin`, 30 minutes), and in addition removes isolated sleep bouts shorter than 30 minutes. The fourth keeps only the main sleep period, the longest sleep bout of each noon-to-noon day of the bridged series. The exact rules are given in Supplementary Table S2.

2.4 Derivation of the sleep/wake series from the raw signal

NHANES 2011–2014 also distributes the raw 80 Hz triaxial acceleration of an ActiGraph GT3X+ worn on the non-dominant wrist for up to nine days [11]. The monitors ran in idle sleep mode (ISM), which outputs a constant value during periods without movement (Supplementary Section 6). The participants of the published sample were placed in a random order with a fixed seed and processed in that order. The analyses use the first 1000 participants, a simple random sample of the published sample, of whom 997 have all four series. For 3 participants SWaN returned no result in two attempts of one hour each, and these participants were excluded. The derivation took 384 hours of processing time over 3.2 days on two laptops and virtual machines of three cloud providers, with equality of the derived series between platforms verified minute by minute (Supplementary Section 7, Table S5).

The signal was processed with `GGIR` 3.3-6 at its defaults [5, 12, 14] (autocalibration, 5-second epochs, non-wear detection, sleep period of each day without a diary, sustained inactivity bouts). Two minute-level series were formed: in `GGIR` SIB (sustained inactivity bouts) a minute is sleep when it lies in the detected sleep period and in a sustained inactivity bout, in `GGIR` SPT (sleep period time) every minute of the sleep period is sleep. The third series came from the public implementation of SWaN [10] (`swan_accel` 1.18), a classifier of 30-second windows developed on NHANES recordings. Its published description matches the NHANES documentation of the released classification [15] (Supplementary Section 6). The rerun used the signal as recorded, without the replacement of the ISM stretches by the preceding 10 seconds of signal that the release documentation describes [16].

Each derived series was placed on the minute index of the released data. In the main analysis a minute entered the SRI only if it was sleep or wake both in the rebuilt and in the derived series, so that all series refer to the same participants, days and minutes. A sensitivity analysis kept each series' own missing minutes (Supplementary Section 7). The SRI of each series was computed with the Phillips formula. Agreement with `PAXPREDM` was summarised as the share of minutes with the same label among the minutes that `PAXPREDM` labels wake or sleep and as Cohen's kappa, overall

and for minutes recorded at least half in ISM. All summaries were also computed after the first 500 participants of the random order (Supplementary Table S6).

2.5 Synthetic recordings and recording gaps

The calculators were also run on five deterministic synthetic recordings with known SRI, a seeded random pattern and a realistic seven-day recording (Supplementary Section 4). The deterministic recordings and the random pattern were presented complete and with one middle day retained as missing minutes. The realistic recording was presented complete, with the middle day retained as missing minutes and with that day deleted from the series.

2.6 Statistical analysis

Agreement between calculators was summarised by correlations, mean paired differences and cross-tabulation of SRI quintiles. For each SRI version the survey-weighted models of the earlier study were refitted on the same 7085-participant frame: the age- and sex-adjusted model, the fully adjusted model (log BMI on SRI plus 13 covariates), the model with a sex $\times$ SRI interaction and the model with sex $\times$ SRI and race/ethnicity $\times$ SRI interactions (Models 1 to 4 of the earlier study). As in the earlier study, SRI values below -6.8, the lower 0.5% point of the SRI of a random sleep/wake pattern, were replaced by draws from that pattern's distribution (8 participants on the released series, one draw per participant where the earlier study used one common draw). The outcome is log BMI without the winsorisation at the 1st and 99th percentiles that the earlier study applied in its Models 2 to 4. With that winsorisation the factors change by at most 0.003 in Models 1 to 3 and 0.011 in Model 4 (`check_bmi_winsorisation.R`). From Model 4, the factor of each race/ethnicity and sex group and the predicted BMI over the SRI range at fixed covariate values were computed as in the earlier study. Effect sizes are multiplication factors $\mathrm{MF} = \exp(\beta_{\mathrm{SRI}}\Delta)$, where Δ is the difference between the median SRI of the top quintile and the median SRI of the bottom quintile, as in the earlier study, with 95% bootstrap confidence intervals (CI, 200 resamples of participants). In each resample, participants were drawn with replacement and the survey design was rebuilt from their

weights, strata and primary sampling units before the weighted model was refitted, as in the earlier study. The earlier study used 500 resamples for Models 1 and 2 and 100 for Models 3 and 4 and reported bootstrap means as point estimates. The re-analysis reports the estimate from the full sample with 200 resamples, which reproduces the published factors to within 0.003 in Models 1 to 3 (Supplementary Table S4) and 0.014 in Model 4 (Section 3.6). The resampling itself is not stratified by the design. Design-based intervals, obtained from the strata and primary sampling units by Taylor linearisation of the weighted estimator, are given for comparison in Supplementary Section 5. The derived series were fitted on the participants of the random sample, with the released classification as reference on the same participants. The ratio of the two factors (derived series to released classification) measures the attenuation of the association. It was formed in each bootstrap resample, so that its interval reflects the pairing of the two estimates on the same participants. Model diagnostics are given in Supplementary Section 5.

A simulation showed the sampling variation of the bootstrap intervals of the multiplication factor [17]. The fitted fully adjusted model on the released series, and separately the fitted sex-interaction model, served as data-generating processes: in each of 20 simulated samples the log BMI of every participant was replaced by its fitted value plus a residual drawn with replacement from the empirical residuals, with SRI, covariates, weights and survey design kept as observed. Each sample was analysed with the full estimation procedure, including the 200-resample bootstrap. Its 95% and 99% intervals were recorded against the factor that generated the data.

A large language model (Claude, Anthropic) assisted with writing the analysis code and the text. The author specified the analyses and checked all code, results, statements and references against the primary sources.

3 Results

3.1 Reproduction of the published SRI

The Phillips formula on the rebuilt series correlated 0.96 with the published values (Supplementary Figure S1). The medians were 61.3 (rebuilt) and 61.5 (published). The fully adjusted top-versus-bottom quintile factor on BMI was 0.955 against the published 0.950 (Supplementary Table S4). The published values themselves were reproduced exactly only with two earlier states of the archived code of the earlier study, which omit the Saturday-to-Sunday day pair and, in 2013–2014, keep partial days (Supplementary Section 3). The code that reproduces them is in the companion repository.

3.2 Epoch-level structure of the released classification

Over the 66.2 million retained minutes, the classification switched between sleep and wake a median of 16 times per participant-day (interquartile range 12 to 22). The median sleep bout lasted 12 minutes and the longest bout of a day a median of 83 minutes. Of all sleep minutes, 80% lay outside the longest bout of the day (Table 1). Sleep was still concentrated at night, with 83 to 85% of the minutes between 02:00 and 05:00 classified as sleep (Supplementary Figure S2). Figure 1 shows seven days of three participants as released and after two of the smoothing rules.

3.3 Calculators on the same input and smoothed input variants

Given the same timeline, `GGIR` and the Phillips formula differed by 0.08 points on average (SD of the difference 2.0, $r = 0.993$) and placed 95% of participants in the same SRI quintile. `sleepreg` differed from the formula by 0.02 to 0.03 points (SD 0.3 and 1.7, $r \geq 0.995$). The agreement held for participants with and without seven consecutive recorded days and for every smoothed input (Table 2, Supplementary Figure S4). The packages differed in the number of participants they scored: `sleepreg` returned no value for 48% of the 7085 participants at its default and for 25% at

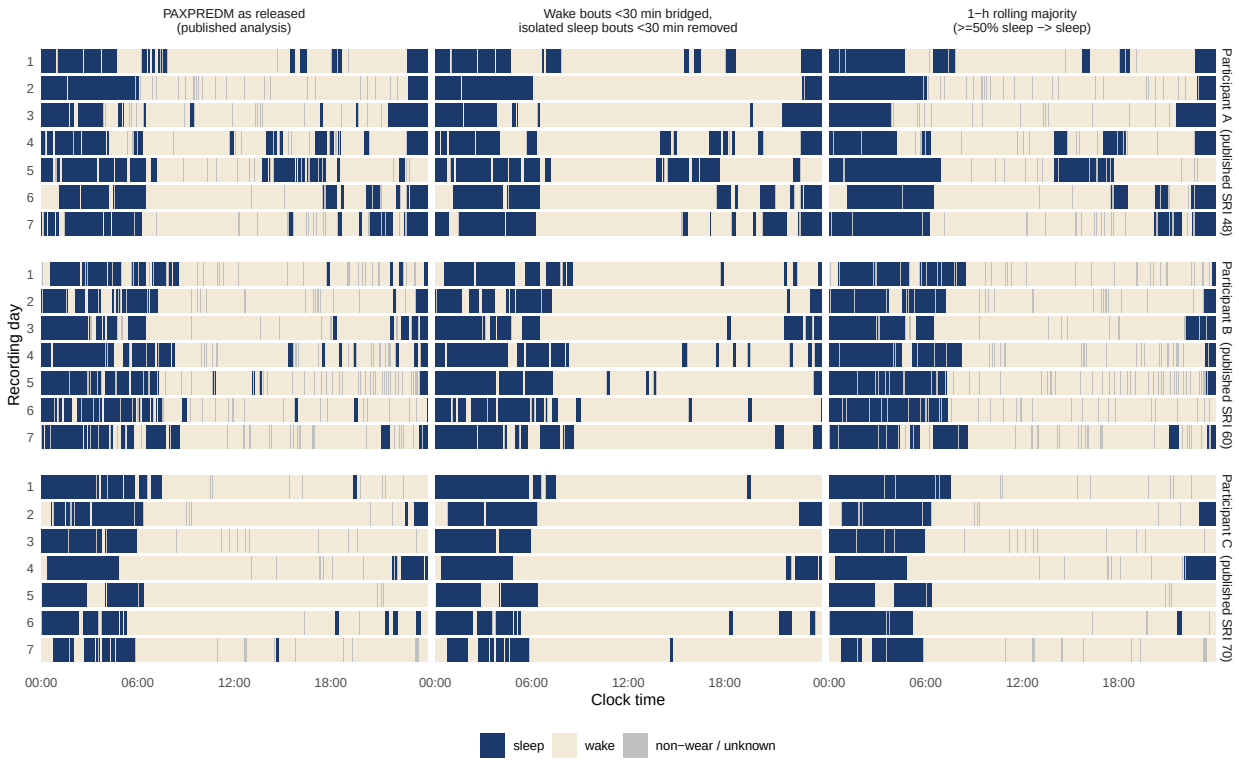


Figure 1: Three participants of the 2013–2014 cycle with seven complete recording days, at the 25th, 50th and 75th percentile of the published SRI, shown as released (left), with awakenings shorter than 30 minutes bridged and isolated sleep bouts shorter than 30 minutes removed (middle), and after the 1-h (61-minute) rolling majority filter (right). Navy, sleep; cream, wake; grey, non-wear or unknown.

Table 1: Epoch-level structure of the released NHANES sleep/wake classification (`PAXPREDM`) in the analytic sample ($n = 7085$, 66.2 million minutes).

Sleep–wake transitions per participant-day, median (IQR)	16 (12–22)
Uninterrupted sleep bout length, median (IQR), min	12 (4–27)
Sleep bouts of 15 minutes or less	59%
Longest sleep bout per day, median (IQR), min	83 (65–108)
Sleep minutes outside the day's longest bout	80%
Sleep bouts per day, median	22
Transitions between 23:00 and 06:00 per night, median (IQR)	8 (5–11)
Wake minutes between 23:00 and 06:00 in bouts of 30 minutes or more	74%
Sleep minutes between 10:00 and 20:00	12%

A bout is a run of consecutive minutes with the same label. Non-wear and unknown minutes end a bout, so that a day has more sleep bouts than half its number of sleep–wake transitions.

Table 2: Median SRI by input variant and calculator (7085 participants, of whom the Phillips formula scores 7076, `GGIR` 7071 to 7073 and `sleepreg` the numbers in parentheses at its default minimum of five comparison days and at three), with the Pearson correlation of each package with the Phillips formula on the same input (for `sleepreg` at three days).

Input	Phillips formula	`GGIR`	r	`sleepreg`, 5 days (n)	`sleepreg`, 3 days (n)	r
Released classification, unsmoothed	61.3	61.4	0.993	61.0 (3673)	61.0 (5342)	0.995
Rolling majority, 1 h	69.1	69.4	0.992	69.1 (3673)	69.0 (5342)	0.994
`GGIR` SRI1 smoothing, 1 h, 80%	69.7	69.8	0.988	69.8 (3673)	69.5 (5342)	0.992
Awakenings <30 min bridged	67.6	67.8	0.994	67.0 (3854)	67.3 (5423)	0.995
Main sleep period only	75.5	75.7	0.990	75.8 (3854)	75.8 (5423)	0.991

three days, because each excluded calendar day removes two of its noon-to-noon days, whereas `GGIR` returned a value for all but 12 to 14 participants, depending on the input. Every smoothing variant raised the SRI. The median of 61.3 on the released series rose to between 67.6 and 69.7 under the rolling majority, the 80% rule and the bridging rule and to 75.5 with the main sleep period alone, while the mean number of transitions fell from 17.7 per day to between 2.1 and 4.5 (Table 2, Supplementary Figure S3 and Section 5).

3.4 Recording gaps

On the synthetic recordings the four implementations returned the known SRI of the five deterministic patterns to within 0.4 points and placed the random pattern less than one point from zero. On the 300 NHANES recordings with seven consecutive days and the fewest missing minutes

they agreed to 0.02 points (Supplementary Section 4, Table S3). With one middle day retained as missing minutes, the formula and `GGIR` agreed (68.75) and `sleepreg` returned no value at its default minimum of comparison days. With the day deleted from the series, the same recording gave 71.94 from `GGIR`, which compared two days 48 h apart, and 35.18 from `sleepreg`, which stores a series as the times of its sleep/wake changes and so carries the last state over the deleted day, and neither package issued a warning (Supplementary Section 4).

3.5 Derivation from the raw signal

On the same participants, days and minutes (the 997 participants of the random sample with all four series), each derived series scored higher than the released classification (Table 3, Figure 2). The mean SRI was 58.1 for `PAXPREDM`, 70.2 for `GGIR` SIB, 75.8 for `GGIR` SPT and 68.3 for SWaN, with mean paired differences from `PAXPREDM` of 12.1 (95% CI 11.3, 12.9), 17.7 (95% CI 16.9, 18.5) and 10.1 (95% CI 9.5, 10.8) points. The released classification changed state a mean of 17.8 times per participant-day and labelled 471 minutes per day as sleep. SWaN, run on the same signal, changed state 3.0 times per day with 365 minutes of sleep. The `GGIR` sleep period changed state 2.0 times with 386 minutes. `GGIR` SIB changed state slightly more often than the released classification (19.5 times per day), because sustained inactivity bouts alternate with movement inside the sleep period. Its SRI was still 12.1 points higher. Minute by minute, the derived series agreed with the released classification in 87.8 to 89.4% of the minutes (kappa 0.74 to 0.77), in 75.4 to 79.9% of the minutes recorded at least half in ISM and in 94.5 to 95.6% of the other minutes. The summaries after the first 500 participants of the random order differed from these by less than a point (Supplementary Table S6, Figure S5). The sensitivity analysis with each series' own missing minutes is given in Supplementary Section 7.

3.6 Association with BMI

With Models 1 to 3 of the earlier study (its Table 3), the fully adjusted top-versus-bottom quintile factor on BMI was 0.955 (95% CI 0.936, 0.976) on the released series and 0.963 to 0.972 on the

Table 3: The released classification and the three series derived from the raw signal on the same participants, days and minutes (random sample, $n = 997$). SRI by the Phillips formula. The difference is the mean paired difference from PAXPREDM with its 95% bootstrap interval. Agreement is the share of minutes with the same label among the minutes that PAXPREDM labels wake or sleep, overall and by whether at least half of the minute was recorded in idle sleep mode (ISM), with Cohen's kappa in parentheses.

	SRI			Transitions	Sleep	Agreement with PAXPREDM, % (kappa)		
Series	Mean (SD)	Median	Difference (95% CI)	per day, mean	min/day, mean	All minutes	ISM	Not ISM
PAXPREDM	58.1 (17.6)	61.3	—	17.8	471	—	—	—
GGIR SIB	70.2 (12.0)	72.2	12.1 (11.3, 12.9)	19.5	336	87.8 (0.74)	75.4 (0.48)	95.6 (0.41)
GGIR SPT	75.8 (15.1)	79.5	17.7 (16.9, 18.5)	2.0	386	88.2 (0.75)	78.4 (0.50)	94.5 (0.42)
SWaN	68.3 (16.0)	71.4	10.1 (9.5, 10.8)	3.0	365	89.4 (0.77)	79.9 (0.54)	95.4 (0.45)

smoothed series (Figure 3, Supplementary Table S4). In women it was 0.922 to 0.946, in men 0.987 to 1.002. Expressed as the percentage difference in BMI between the top and bottom SRI quintile, the fully adjusted factor was -4.5% on the released series and -2.8% to -3.7% after smoothing. The factors from GGIR differed from those of the formula by at most 0.002, those from sleepreg were 0.009 to 0.020 lower because of its smaller sample (Supplementary Section 5).

In Model 4, the factors of the twelve race/ethnicity and sex groups on the published SRI differed from those of the earlier study by at most 0.014 (its Table 3 reports bootstrap means). The pattern persisted under every input: in each race/ethnicity group, women had a lower factor than men in 36 of 36 comparisons (six groups, six inputs). Among the five larger race/ethnicity groups the lowest factors were those of non-Hispanic White women, 0.912 (95% CI 0.887, 0.945) on the released series and 0.934 (95% CI 0.906, 0.963) with the main-sleep-period input. The smallest group, other race/ethnicity including multiracial (111 women), had lower point estimates with wide intervals (0.887 (95% CI 0.797, 0.988) on the released series). The highest factors were those of Mexican American and non-Hispanic Black men (1.034 and 1.035 on the released series). Smoothing moved the group factors by -0.020 to $+0.030$ (Supplementary Section 8, Table S8, Figures S6 and S7). GGIR agreed with the formula to 0.013 or less in every group and input. sleepreg differed by up to 0.064 in the smallest groups because of its smaller sample.

On the 997 participants of the random sample, the released classification gave a fully adjusted factor of 0.952 (95% CI 0.907, 0.996). The derived series gave 1.004 (95% CI 0.955, 1.041) for

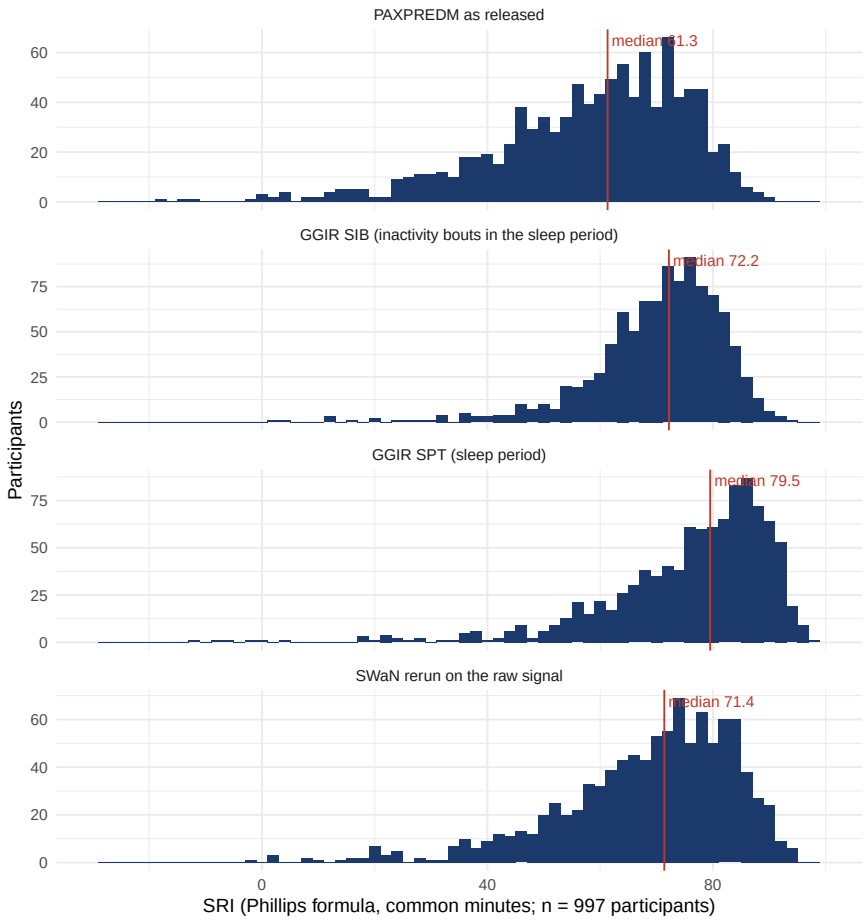


Figure 2: Distribution of the SRI (Phillips formula on the minutes labelled in both series) for the released classification and the three series derived from the raw signal in the random sample ($n = 997$), with medians.

GGIR SIB, 0.999 (95% CI 0.952, 1.037) for GGIR SPT and 0.993 (95% CI 0.951, 1.025) for SWaN (Table 4). On the same participants, the four smoothed versions of the released classification gave 0.969 to 0.979, above the factor of the released classification as in the published sample. Divided by the factor of the released classification in the same bootstrap resamples, the factors of the derived series gave ratios of 1.055 (95% CI 1.014, 1.096), 1.049 (95% CI 1.012, 1.084) and 1.043 (95% CI 1.010, 1.077), where a ratio above 1 means a weaker inverse association. The weaker association is not an effect of the narrower quintile contrast of the derived series: per 10 SRI points, the BMI change was -1.1% for the released classification and -0.2% to $+0.2\%$ for the derived series (Supplementary Section 7). The sex pattern of the earlier study appeared in every series, with point

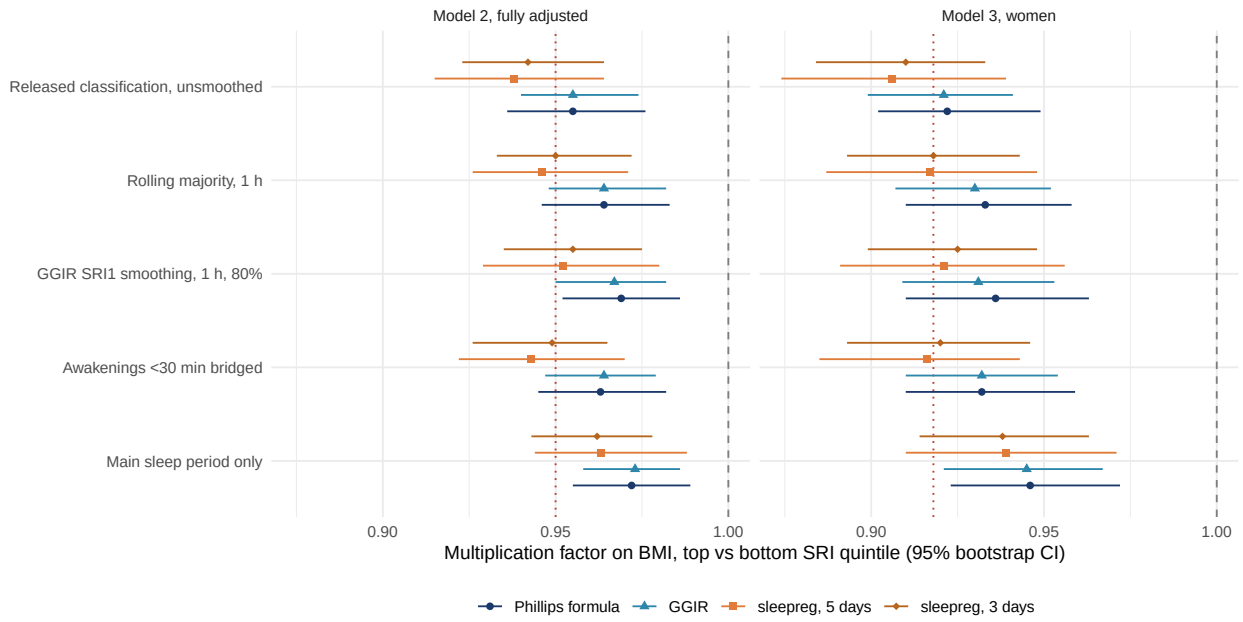


Figure 3: SRI–BMI multiplication factor (top versus bottom SRI quintile) by input variant and calculator, with 95% bootstrap intervals; left, fully adjusted model; right, women. The dotted red line marks the published value and the dashed line no association. `sleepreg` estimates rest on the participants for whom it returned a score (5 days: $n = 3673$ to 3854; 3 days: 5342 to 5423). All inputs are the released classification and its smoothed versions in the published sample. The series derived anew from the raw signal are compared with these inputs on the same participants in Table 4. The estimates are listed in Supplementary Table S4.

estimates below one in women and above one in men (Table 4). The sensitivity analysis with each series' own missing minutes gave factors that differed from these by at most 0.01 (Supplementary Table S7). Model 4 on the random sample, with 11 to 226 participants per race/ethnicity and sex group, kept the sex pattern in every series (women below men in all six groups for each of the four series) with wide intervals (widths 0.13 to 0.88, Supplementary Table S9).

Figure 4 shows the intervals of the 20 samples simulated from the fitted models. With the fully adjusted factor fixed at 0.955, the point estimates ranged from 0.936 to 0.964 and no 95% interval reached above 0.981. The 95% intervals contained the true factor in 18 of 20 samples and the 99% intervals in 20. For women (true factor 0.922), the point estimates ranged from 0.894 to 0.942. The 95% intervals contained the true factor in 18 of 20 samples, the 99% intervals in 19.

Table 4: SRI–BMI multiplication factor (top versus bottom SRI quintile) on the same 997 participants (random sample) for the released classification, its four smoothed versions (the inputs of Figure 3) and the three series derived anew from the raw signal (survey-weighted Models 1 to 3 of the earlier study, its Table 3; 95% bootstrap intervals). ΔSRI is the median SRI difference between the top and bottom quintile of each series. Model 1 adjusts for age and sex, Model 2 is fully adjusted, Model 3 adds a sex × SRI interaction. The last column is the ratio of the Model 2 factor to that of the unsmoothed released classification, formed in the same bootstrap resamples. Model 4, which gives one factor per race/ethnicity and sex group, is in Supplementary Table S9.

Series	ΔSRI	Model 1 (age, sex)	Model 2 (fully adjusted)	Model 3, women	Model 3, men	Ratio to released (Model 2)
Released classification (Phillips formula) and its smoothed versions						
Released classification, unsmoothed	42.9	0.920 (0.871, 0.968)	0.952 (0.907, 0.996)	0.888 (0.819, 0.951)	1.014 (0.957, 1.067)	—
Rolling majority, 1 h	44.4	0.938 (0.895, 0.985)	0.970 (0.925, 1.010)	0.909 (0.842, 0.973)	1.026 (0.968, 1.073)	1.019 (1.007, 1.030)
GGIR SRI1 smoothing, 1 h, 80%	37.4	0.935 (0.889, 0.983)	0.969 (0.927, 1.015)	0.918 (0.850, 0.989)	1.020 (0.966, 1.067)	1.018 (0.997, 1.045)
Awakenings <30 min bridged	45.9	0.938 (0.894, 0.981)	0.970 (0.926, 1.013)	0.910 (0.844, 0.969)	1.027 (0.966, 1.080)	1.019 (1.007, 1.030)
Main sleep period only	39.1	0.952 (0.906, 0.996)	0.979 (0.936, 1.023)	0.945 (0.885, 1.005)	1.012 (0.947, 1.068)	1.028 (0.996, 1.068)
Series derived anew from the raw signal						
GGIR SIB	26.5	0.969 (0.925, 1.006)	1.004 (0.955, 1.041)	0.967 (0.904, 1.016)	1.050 (0.987, 1.106)	1.055 (1.014, 1.096)
GGIR SPT	33.6	0.964 (0.926, 1.006)	0.999 (0.952, 1.037)	0.973 (0.917, 1.021)	1.028 (0.966, 1.083)	1.049 (1.012, 1.084)
SWaN	37.9	0.957 (0.913, 0.994)	0.993 (0.951, 1.025)	0.956 (0.900, 1.012)	1.028 (0.967, 1.070)	1.043 (1.010, 1.077)

4 Discussion

The sleep/wake classification released with NHANES 2011–2014 is highly fragmented. Any SRI computed from it, by any calculator, is the SRI of a fragmented series. The unweighted median of the published values, 61.5, should be read in this light. Smoothing raised the median to between 67.6 and 75.5, closer to the medians of 74.7 and 81 reported for epoch series derived with a sleep-period detection step [3, 9]. The series derived anew from the raw signal of a random sample scored 10.1 to 17.7 points above the released classification. The calculators themselves agreed on the same series. Under every smoothed version and every calculator the inverse SRI–BMI association kept its direction. With the Phillips formula and `GGIR` it lost a fifth to two fifths of its size. With the series derived anew from the raw signal the point estimates moved close to no association: the fully adjusted multiplication factor (Model 2 in Table 4) moved from 0.952 to between 0.993 and 1.004, where 1 means no association, with 95% intervals that reach from a small inverse association (lower limits 0.951 to 0.955, just above the published 0.950) to a small positive one (upper limits 1.025 to 1.041).

4.1 Fragmentation of the input and the level of the SRI

The released series shows a main sleep period at the expected time of day, cut into short pieces by brief awakenings and unknown epochs, with scattered short sleep bouts during the day (Figure 1). Fragmentation lowers the SRI because the index counts the epoch pairs 24 h apart that are in the same state. A brief awakening or an unknown minute inside the sleep period that does not recur at the same clock time on the next night turns matching pairs into mismatches, as does a short daytime sleep bout on one day only. The classifier behind `PAXPREDM` [10] labels 30-second windows and returns no consolidated sleep period. The post-processing described in the survey documentation is limited to minimum durations of 3 minutes for wake and 10 minutes for sleep and non-wear and to an orientation rule [16]. The present data cannot decide which smoothing is appropriate. The rolling majority and the bridging rule leave total sleep almost unchanged, the 80% rule removes

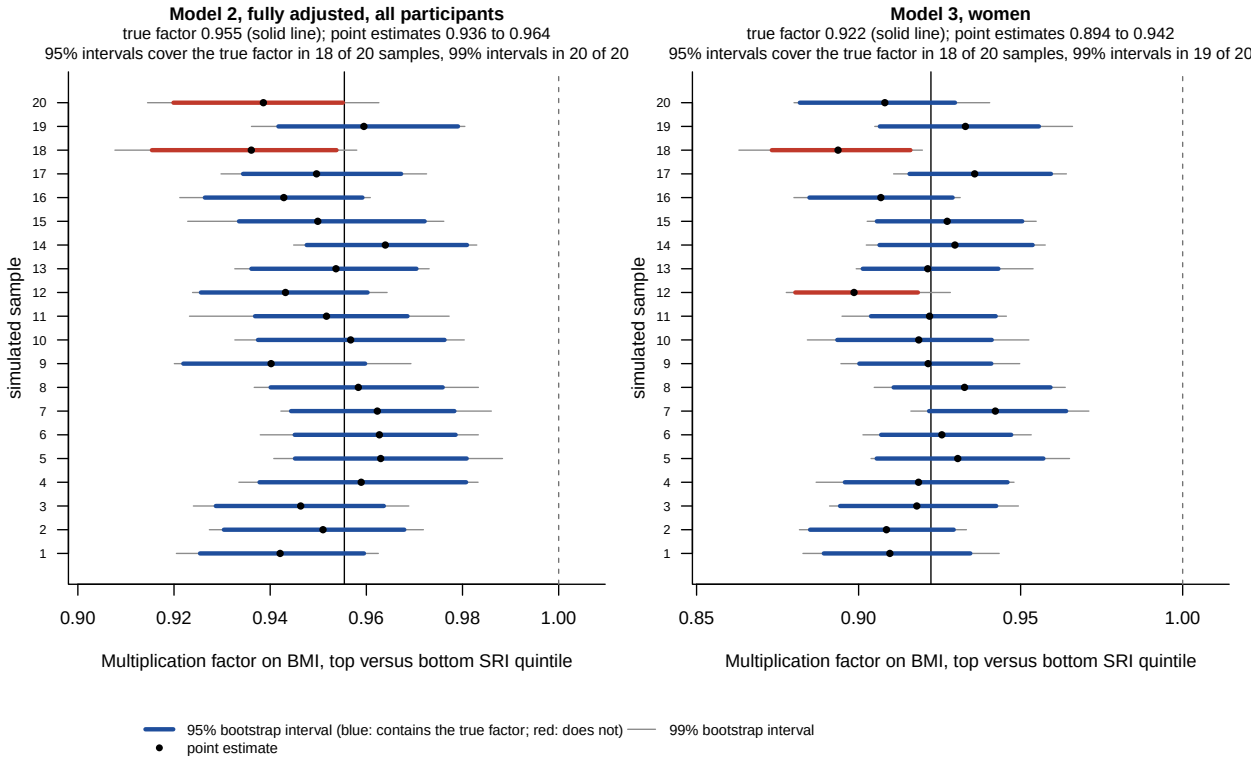


Figure 4: Twenty samples simulated from the fitted models on the released series, with the true multiplication factor known (solid line): point estimates with 95% (thick) and 99% (thin) bootstrap intervals of the SRI–BMI factor, top versus bottom SRI quintile; left, fully adjusted model; right, women. The dashed grey line marks no association. A 95% interval drawn in red does not contain the true factor.

approximately 100 minutes of sleep per day, and the main-sleep-period rule discards the naps that the SRI as defined by Phillips et al. [1] includes. The four variants do show that the smoothing decision is at least as important a reporting item as the calculator: it moved the level by up to 14 points and the effect size by up to two fifths. The spread of reported NHANES SRIs, from 78.1 [7] to 63.0 [8], is about as wide, and neither analysis published its code.

4.2 Calculators, gaps and defaults

Czeisler et al. [20] found score differences of approximately 20 points between `GGIR` and `sleepreg` in the UK Biobank, where mortality studies based on `sleepreg` [18] and on `GGIR` [19] differ in effect size. In their comparison the packages shared `GGIR`'s epoch states, but `sleepreg` then

re-handled naps and awakenings with its own rules. In the present analysis that re-handling did not run, the series was identical down to the representation of its gaps, and the packages agreed. The two findings are compatible if the divergence in the UK Biobank arises in the derivation of the sleep/wake series, since the arithmetic of the index is the same in all packages. The packages differed at gaps in the data, where the index compares each epoch with the epoch exactly 24 h later and has no pair when either is missing. A day deleted from a series instead of marked as missing is bridged without warning: `GGIR` pairs the date before the gap with the next date present, and `sleepreg` extends the last state over the gap. `sleepreg` also blanks any noon-to-noon day with more than 6 h missing, so that a single excluded calendar day costs two of its comparison days. With the default minimum of five, no NHANES participant with a gap received a score. The 48% lost in this way were selected by wear behaviour, since only participants with seven consecutive recorded days remain, and are not a random subset. None of these behaviours is an error in packages written for continuous raw recordings, but each is undocumented for a pre-classified series with excluded days, which is how a survey such as NHANES is used.

4.3 Derivation from the raw signal and the association with BMI

Run on the raw signal of the same participants, the public code of SWaN produced a series with 3.0 transitions per day where the released classification has 17.8. The two series agree in 89.4% of the minutes. The disagreement is concentrated in the minutes recorded in idle sleep mode (79.9% against 95.4%). Two differences between the runs can account for this: the released classification was produced with a procedure that replaces the ISM stretches by the 10 seconds of signal that precede them [16], whereas the rerun used the signal as recorded, and the version of the code behind the release is not documented. The fragmentation of the released series therefore does not follow from the public SWaN code. It belongs to the run that produced the release. The concentration of the disagreement in ISM minutes is consistent with a role of the ISM replacement, although the `GGIR` series, which involve no such replacement either, show the same concentration.

The association with BMI weakened with the derived series more than with the smoothed

variants: on the same 997 participants, the smoothed versions of the released classification gave fully adjusted factors of 0.969 to 0.979 and the derived series 0.993 to 1.004 (Table 4). The ratios of the fully adjusted factors to that of the released classification were 1.043 to 1.055. The factors themselves, 0.993 to 1.004, are compatible with no association as well as with a small inverse or a small positive one. Two readings remain open. If the derived series measure the construct with less noise, the published association was inflated by an error in the released series that is itself related to adiposity (daytime sleep, movement during sleep or non-wear labelled as sleep). Random error alone would have attenuated it, as regression dilution predicts. If the fragmentation carries information about sleep that is itself related to BMI, the derived series remove it. The present data cannot separate the two. They show that the size of the association depends on the processing pipeline more than on the calculator, with the fully adjusted factor, expressed as the percentage difference in BMI between the top and bottom quintile, from approximately -5% for the released classification (-4.8% on the same participants, -4.5% in the published sample) to approximately zero for the series derived anew from the signal (Table 4, Supplementary Table S4). A reader who classifies results by whether the 95% interval includes 1 would reach opposite conclusions from the same participants, an inverse association with the released classification and none with the derived series (Table 4). This reading is the one that Wasserstein et al. [23] advise against: an interval shows the effect sizes most compatible with the data under the model. Whether it includes 1 is not a result.

For the earlier study itself, the rebuilt series moved the fully adjusted estimate from 0.950 to 0.955 and left the sex pattern unchanged. The sex and race/ethnicity pattern of its Model 4 (its Figure 2) persisted under every input (Supplementary Table S8, Figures S6 and S7). Its Methods describe the final state of its code and not the computation behind its reported values. This difference does not change its conclusions, but it belongs beside its reported SRI values (Supplementary Section 3).

4.4 Limitations

The series was rebuilt from the public files with the filters as published, whereas the published scores came from earlier states of the processing code. The same-input baseline is hence the formula on the rebuilt series. The two give effect sizes that differ by 0.005. No reference measure of sleep exists for these participants, so that the agreement between derivations is a concordance between algorithms. The sleep detection of `GGIR` was developed on other monitors [5, 12] and applied to recordings in idle sleep mode. The derivation covers a random sample of 997 participants, about one seventh of the published sample, with correspondingly wider intervals. The summaries after the first 500 participants differed by less than a point from those after 1000 (Supplementary Table S6). SWaN was run without the ISM replacement that the survey documentation describes and with the public version of its code, whose relation to the version behind the release is unknown. The autocalibration of `GGIR` depends on the floating-point precision of the processor. The derived series were computed on five platforms. For the 80 participants processed both on the Apple platform and on another machine, the sleep/wake minute labels of all three series were identical (Supplementary Section 7, Table S5). The smoothing rules are simple post-processing steps without validation as sleep-period detectors. The main-sleep-period variant changes the construct by removing naps.

4.5 Reporting items for pre-classified series

The RIRI statement asks for the data processing tool, the quality control and the additional configurations of an SRI analysis [20]. For a pre-classified series, five details fill these items. (1) How the epoch-level series was derived (classifier, version, handling of the monitor's idle sleep mode), whether and how it was smoothed, and its fragmentation (transitions per day, bout lengths). (2) How excluded minutes and days were passed to the calculator, as missing values or as deletions, and whether the package's own inclusion criterion was applied to its day pairs. (3) The minimum-day rule and the analytic sample it leaves. (4) Effect sizes with compatibility intervals for each processing choice that was tried. (5) The state of the processing code that produced the reported values, archived with them. Item (4) asks for the interval as a range of compatible values. Figure 4

shows why: in 20 samples simulated from the fitted model, the point estimates moved between 0.936 and 0.964 and the interval ends by more, while the factor that generated the data stayed at 0.955. One interval from one sample is one draw from this variation [17, 21]. The completed RIRI checklist of the present study is Supplementary Table S1. A World Sleep Society specification for consumer sleep trackers [22] did not adopt the SRI, citing the lack of a standard computation. It specifies instead that the 30-second sleep/wake/unknown epoch array should be retained. Such an array needs to carry the classification, the smoothing applied to it, the inclusion filters and the position of its gaps.

5 Conclusions

The level of the SRI in NHANES 2011–2014 depended on the sleep/wake input and only marginally on the calculator. Given the same series, the Phillips formula, `GGIR` and `sleepreg` agreed to 0.1 points. Smoothing the released classification raised the median SRI from 61 to 68–76. The three series derived anew from the raw signal scored 10 to 18 points higher than the release, and the released classification is more fragmented than a classification of the same signal with the public code of the same algorithm. The inverse SRI–BMI association kept its direction under every smoothed input and, with the Phillips formula and `GGIR`, lost a fifth to two fifths of its size. With the series derived anew from the raw signal the point estimates moved close to no association, with intervals reaching from a small inverse association near the published value to a small positive one. The packages differ in how they treat a recording gap. A day deleted instead of marked as missing changes the result without warning. Studies of the SRI should report the derivation and smoothing of the series, the treatment of gaps, the minimum-day rule and the remaining sample. The processing code should be archived with the results.

List of abbreviations

BMI body mass index

CI confidence interval

IQR interquartile range

ISM idle sleep mode

MF multiplication factor

NCHS National Center for Health Statistics

NHANES National Health and Nutrition Examination Survey

PAXPREDM predicted sleep/wake state in the NHANES minute-level accelerometry files

RIRI Reporting Items for Regularity Indices

SD standard deviation

SIB sustained inactivity bout

SPT sleep period time

SRI Sleep Regularity Index

SWaN sleep-wear, wake-wear and nonwear classifier

Declarations

Ethics approval and consent to participate. This is a secondary analysis of de-identified, publicly available NHANES 2011–2014 data. The NHANES protocol was approved by the National Center for Health Statistics (NCHS) Research Ethics Review Board, and all participants provided written informed consent at the time of the survey. No further ethics approval was required for this re-analysis.

Consent for publication. Not applicable. The study uses de-identified public data and reports no individually identifiable information.

Availability of data and materials. All data are NHANES 2011–2012 and 2013–2014 public-use files of the US National Center for Health Statistics (`https://wwwn.cdc.gov/nchs/nhanes/`): the Physical Activity Monitor minute and header files (`PAXMIN`, `PAXHD`), the raw 80 Hz acceleration files (`PAX80_G` and `PAX80_H`, one archive per participant, 2.45 TB in total) and the covariate files of the earlier study. The companion repository (`https://github.com/jdegenfellner/SRI_BMI_NHANES_2026_reanalysis_public`) contains the code of every step, the analytic covariate data set, the derived SRI values of every calculator and input variant, the minute-level sleep/wake series of every derivation, the container definition of the cloud runs and the code that reproduces the published SRI values exactly. A frozen copy with the recoded minute files and all intermediate series is archived at Zenodo (`https://doi.org/10.5281/zenodo.23482613`), so that every table and figure can be recomputed from the archived files without any download (run order in `code/revision_2026-10/README.md`).

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Use of artificial intelligence. A large language model (Claude, Anthropic) assisted with drafting and language editing of the manuscript and with writing and executing the analysis code, as stated in the Methods. The author takes full responsibility for the content.

References

1. Phillips AJK, Clerx WM, O'Brien CS, Sano A, Barger LK, Picard RW, Lockley SW, Klerman EB, Czeisler CA. Irregular sleep/wake patterns are associated with poorer academic performance and delayed circadian and sleep/wake timing. Sci Rep. 2017;7:3216. https://doi.org/10.1038/s41598-017-03171-4

2. Fischer D, Klerman EB, Phillips AJK. Measuring sleep regularity: theoretical properties and practical usage of existing metrics. Sleep. 2021;44(10):zsab103. https://doi.org/10.1093/sleep/zsab103

3. Windred DP, Jones SE, Russell A, Burns AC, Chan P, Weedon MN, Rutter MK, Olivier P, Vetter C, Saxena R, Lane JM, Cain SW, Phillips AJK. Objective assessment of sleep regularity in 60,000 UK Biobank participants using an open-source package. Sleep. 2021;44(12):zsab254. https://doi.org/10.1093/sleep/zsab254

4. Migueles JH, Rowlands AV, Huber F, Sabia S, van Hees VT. GGIR: a research community-driven open source R package for generating physical activity and sleep outcomes from multi-day raw accelerometer data. J Meas Phys Behav. 2019;2(3):188–96. https://doi.org/10.1123/jmpb.2018-0063

5. van Hees VT, Sabia S, Anderson KN, Denton SJ, Oliver J, Catt M, Abell JG, Kivimäki M, Trenell MI, Singh-Manoux A. A novel, open access method to assess sleep duration using a wrist-worn accelerometer. PLoS One. 2015;10(11):e0142533. https://doi.org/10.1371/journal.pone.0142533

6. Degenfellner J, Schernhammer ES, Strohmaier S. Sex- and ethnic differences in the cross-sectional association between sleep regularity and obesity among US adults, NHANES 2011–2014. J Biol Rhythms. 2026;41(2):205–18. https://doi.org/10.1177/07487304251391267

7. Wang X, Xu Y, Li X, Mansuri A, McCall WV, Liu Y. Day-to-day deviations in sleep parameters and biological aging: findings from the NHANES 2011–2014. Sleep Health. 2023;9(6):940–6. https://doi.org/10.1016/j.sleh.2023.07.018

8. Yang L, Wallen G, Barb J, Maki K, Farmer N. 0863 Sleep regularity and cardiovascular health among U.S. adults [conference abstract]. Sleep. 2024;47(Suppl 1):A370–1. https://doi.org/10.1093/sleep/zsae067.0863
9. Lunsford-Avery JR, Engelhard MM, Navar AM, Kollins SH. Validation of the sleep regularity index in older adults and associations with cardiometabolic risk. Sci Rep. 2018;8:14158. https://doi.org/10.1038/s41598-018-32402-5
10. Thapa-Chhetry B, Arguello DJ, John D, Intille S. Detecting sleep and nonwear in 24-h wrist accelerometer data from the National Health and Nutrition Examination Survey. Med Sci Sports Exerc. 2022;54(11):1936–46. https://doi.org/10.1249/MSS.0000000000002973
11. National Center for Health Statistics. National Health and Nutrition Examination Survey 2011–2012 data documentation, codebook, and frequencies: physical activity monitor, raw data 80 Hz (PAX80_G). Hyattsville (MD): Centers for Disease Control and Prevention; 2022. https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/2011/DataFiles/PAX80_G.htm
12. van Hees VT, Sabia S, Jones SE, Wood AR, Anderson KN, Kivimäki M, Frayling TM, Pack AI, Bucan M, Trenell MI, Mazzotti DR, Gehrman PR, Singh-Manoux A, Weedon MN. Estimating sleep parameters using an accelerometer without sleep diary. Sci Rep. 2018;8:12975. https://doi.org/10.1038/s41598-018-31266-z
13. Hammad G, Reyt M, Beliy N, Baillet M, Deantoni M, Lesoinne A, Muto V, Schmidt C. pyActigraphy: open-source python package for actigraphy data visualization and analysis. PLoS Comput Biol. 2021;17(10):e1009514. https://doi.org/10.1371/journal.pcbi.1009514
14. van Hees VT, Fang Z, Langford J, Assah F, Mohammad A, da Silva ICM, Trenell MI, White T, Wareham NJ, Brage S. Autocalibration of accelerometer data for free-living physical activity assessment using local gravity and temperature: an evaluation on four continents. J Appl Physiol (1985). 2014;117(7):738–44. https://doi.org/10.1152/japplphysiol.00421.2014
15. Koffman L, Muschelli J. Minute level step counts and physical activity data from the National

Health and Nutrition Examination Survey (NHANES) 2011–2014 (version 1.0.2) [data set]. PhysioNet; 2026. https://doi.org/10.13026/d7sw-f662

16. National Center for Health Statistics. National Health and Nutrition Examination Survey 2011–2012 data documentation, codebook, and frequencies: physical activity monitor, minute (PAXMIN_G). Hyattsville (MD): Centers for Disease Control and Prevention; 2022. https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/2011/DataFiles/PAXMIN_G.htm
17. Cumming G. The new statistics: why and how. Psychol Sci. 2014;25(1):7–29. https://doi.org/10.1177/0956797613504966
18. Windred DP, Burns AC, Lane JM, Saxena R, Rutter MK, Cain SW, Phillips AJK. Sleep regularity is a stronger predictor of mortality risk than sleep duration: a prospective cohort study. Sleep. 2024;47(1):zsad253. https://doi.org/10.1093/sleep/zsad253
19. Cribb L, Sha R, Yiallourou S, Grima NA, Cavuoto M, Baril AA, Pase MP. Sleep regularity and mortality: a prospective analysis in the UK Biobank. eLife. 2023;12:RP88359. https://doi.org/10.7554/eLife.88359
20. Czeisler MÉ, Windred DP, Phillips AJK, Sabia S, van Hees VT, Rutter MK, Cain SW. Comparison of Sleep Regularity Index scores calculated by open-source packages and implications for outcomes research: rationale and design of the RIRI statement. Sleep. 2026;49(4):zsaf299. https://doi.org/10.1093/sleep/zsaf299
21. Greenland S, Senn SJ, Rothman KJ, Carlin JB, Poole C, Goodman SN, Altman DG. Statistical tests, P values, confidence intervals, and power: a guide to misinterpretations. Eur J Epidemiol. 2016;31(4):337–50. https://doi.org/10.1007/s10654-016-0149-3
22. Chee MWL, Baumert M, Drummond SPA, Scott H, Goldstein CA, de Zambotti M, Heneghan C, Leota J, Hao S, Cimesa L, Cellini N, Overeem S, Baron KG, Sharafkhaneh A, Penzel T, Kushida CA. Standardizing wearable sleep data for clinical, research, and public health use: a World Sleep Society specification. Sleep Med. 2026;109310. Advance online publication. https://doi.org/10.1016/j.sleep.2026.109310
23. Wasserstein RL, Schirm AL, Lazar NA. Moving to a world beyond "$p < 0.05$". Am Stat.

2019;73(sup1):1–19. https://doi.org/10.1080/00031305.2019.1583913